



Ad Copilot

An AI Co-Pilot for Performance Marketing Teams

Product Portfolio Project — Zero-to-One AI Product for Performance Marketing

Case Study · PRD · User Stories · AI Workflow Design · Metrics & Attribution Spec · Architecture · Release Plan

How This Deck Is Organized

Five sections, each built to answer a specific part of the job requirements — and each useful to a different reader.

01



Problem & Zero-to-One Build

Audience: Hiring Manager, Growth Lead

02



Managing AI Products

Audience: Hiring Manager, ML/Eng

03



Performance Marketing Metrics

Audience: Marketing / Growth Stakeholders

04



PM Craft: PRDs, Stories, Release Plans

Audience: PM Peers, Hiring Manager

05



Engineering Collaboration

Audience: Engineering Lead

01



Problem & Zero-to-One Build

Why this product needed to exist, and who it's for

Audience: Hiring Manager, Growth Lead

The Problem, and Who Feels It



Two Compounding Problems

Creative decay outpaces production. Winning ads fatigue in 1-3 weeks; small teams can't hand-write replacements fast enough.

Performance signals are caught too late. CPA/ROAS drift is typically noticed only after days of wasted spend.

The real gap isn't AI capability — it's trust. Creative tools and bid automation exist separately; no layer combines both with a transparent trust model.



Primary Persona

In-House Performance Marketer

- Manages \$50K-\$500K/mo in paid social & search

Team of 1-4, no dedicated data scientist

Distrusts platform "black box" automation from past experience

Will not adopt anything that can spend money invisibly

Biggest time sink is production, not analysis

The Solution: Three Connected Capabilities

Ad Copilot generates creative, monitors performance, and routes every action through a confidence-based human review lane.



Creative Generation

Generates ad copy variants from the product feed and past top performers, each scored for predicted confidence.



Performance Monitoring

Watches live ROAS/CPA/CTR against targets and proposes budget or bid changes when performance drifts.



Human Review Lanes

Every AI action lands in Auto-apply, Suggest, or Escalate — proportional to confidence and financial risk.

Key Zero-to-One Decisions

The tradeoffs that shaped scope — the parts of a 0-to-1 build that don't show up in a feature list.

Default to "Suggest," not "Auto-apply," in v1

Even high-confidence actions stay human-approved until trust is earned per account.

Cost controls are a v1 requirement

Usage-scaled LLM spend can outpace the customer's own ad budget if left to v2.

No custom attribution engine

Platform-native attribution preserves credibility with numbers marketers already trust.

One north-star metric, not a dashboard

Incremental ROAS from accepted suggestions — not engagement with the tool.

02



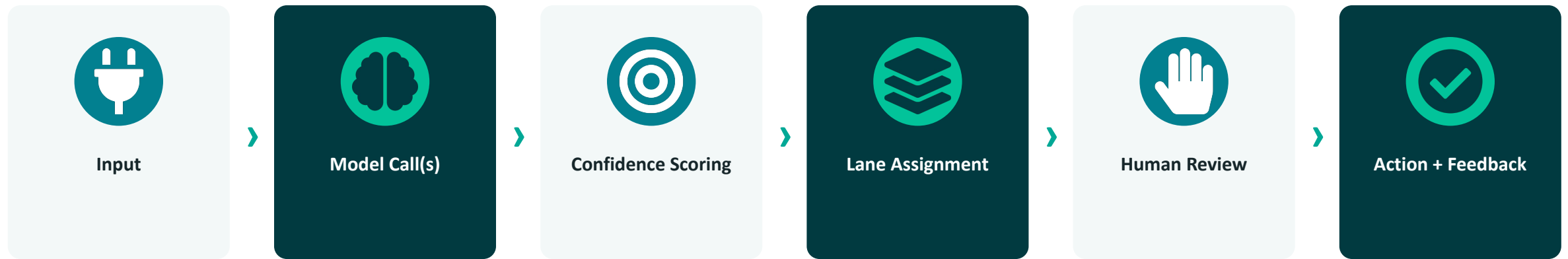
Managing AI Products

Workflows, model limitations, prompt behavior, cost controls, and human review

Audience: Hiring Manager, ML/Eng

The AI Workflow, End to End

Every AI-driven action — creative or budget — follows the same shape.



The feedback loop is the mechanism: every human decision recalibrates future confidence scoring — trust is earned, not assumed.

Confidence-Based Review Lanes

Autonomy is proportional to confidence, blast radius, and account trust tier — never a blanket setting.



Auto-apply

**High confidence + low blast radius +
established trust tier**

*e.g., a 5% budget nudge on a campaign with 60
days of stable, accepted history*



Suggest

**Medium confidence, or above blast-radius cap,
or newer account**

*Most v1 volume lands here by design — this is
where trust gets built*



Escalate

**Low confidence, thin data, or
compliance-sensitive action**

*e.g., a recommendation based on fewer than 100
conversions of data*

Model Limitations & Cost Controls

Designing around what the model gets wrong, and what it costs to run at scale.



Known Model Limitations

- Noisy per-item confidence — never shown raw; smoothed and bucketed (High/Med/Low).

Hallucinated specificity — reasoning is constrained to only reference input fields, no invented claims.

Prompt injection via product feed — feed content treated as untrusted, isolated in the prompt structure.



Cost Control Mechanisms

- Tiered model routing — cheap model for first-pass at volume, strong model only on top-N candidates.

Per-account monthly generation cap, enforced before the call is made — not billed-then-refunded.

Caching on unchanged feed inputs, keyed to prompt-version so updates still invalidate correctly.

Circuit breaker on provider latency/errors — degrades gracefully instead of retrying into an incident.

03



Performance Marketing Metrics

ROAS, CPA, CPL, CPC, CTR, attribution, and conversion tracking as product signals

Audience: Marketing / Growth Stakeholders

Metrics as Product Signals, Not Just Vocabulary

All five metrics are sourced platform-native — credibility depends on matching what marketers already see in Ads Manager.

Metric	Definition	Used For
ROAS	Attributed revenue ÷ ad spend	Primary drift target, e-commerce accounts
CPA	Ad spend ÷ attributed conversions	Primary drift target, lead-gen accounts
CPL	Ad spend ÷ attributed leads	Secondary target, multi-stage funnels
CPC	Ad spend ÷ clicks	Leading indicator ahead of CPA/ROAS shifts
CTR	Clicks ÷ impressions	Primary creative-fatigue detection signal

Attribution & Data Quality Handling



Platform-native attribution, deliberately — not a custom multi-touch model

A custom attribution layer that disagrees with the platform's own dashboard is an instant credibility problem for a performance marketer.

iOS / App Tracking Transparency

High iOS conversion share can mean modeled, not fully observed, conversions — the platform's modeled-vs-observed flag is surfaced, not hidden.

View-through vs. click-through

Reported separately since blending inflates apparent CPA strength; marketer chooses whether to include view-through in their target.

Provisional / sampled data

Platform APIs report provisional numbers that shift for 24-72 hours; any recommendation on provisional data is annotated as such.

04



PM Craft: PRDs, Stories, Release Plans

Detailed requirements, acceptance criteria, and phased rollout planning

Audience: PM Peers, Hiring Manager

From Requirements to Acceptance Criteria



PRD Snapshot

Goal: >40% suggestion acceptance by month 3

Non-goal: no autonomous spend above threshold, no custom attribution, text-only creative in v1

15 functional requirements across creative gen, monitoring, review lanes, and cost controls

4 non-functional requirements covering rate-limit handling, rollback speed, and auditability



Example: US-9

"As a performance marketer, I want to approve, reject, or edit a queued recommendation so that I retain control over budget-impacting actions."

Acceptance criteria (Given/When/Then):

- Given a Suggest-lane item, when I click Approve, then the action executes and is logged.

Given a Suggest-lane item, when I click Edit, then I can change the proposed value before approving.

Given I reject an item, when I do so, then I can add a reason, stored to improve future scoring.

Release Plan: Earning Autonomy in Phases

Each phase is gated on trust and quality signals, not a calendar date.

Phase 1

MVP — Meta Only

Creative generation + Suggest-lane review. 5-10 design partners.

Exit: $\geq 40\%$ acceptance, sustained 4+ weeks



Phase 2

Multi-Platform

Add Google Ads + budget/bid recommendations (still Suggest-only).

Exit: budget-rec acceptance reaches creative-rec parity



Phase 3

Auto-Apply

Opt-in autonomy for trust-tier-eligible accounts, hard ceilings enforced.

Success: stable/low rollback rate, ROAS $\geq$ Suggest-lane history

05



Engineering Collaboration

Architecture, API integrations, data flows, and open technical questions

Audience: Engineering Lead

System Architecture at a Glance



Ingestion Service

Hourly pulls from Meta/Google Ads APIs; OAuth refresh, rate-limit backoff



Drift Detection

Compares live metrics to targets, generates recommendation objects



Creative Gen + Scoring

Tiered LLM routing, confidence scoring, lane assignment



Review Queue

Marketer approve/edit/reject surface; source of the feedback loop



Action Executor

Only component with write access to live ad platform campaigns



Audit Log

Immutable record of every action, across all lanes, for rollback

Open with engineering: Google's tighter mutate rate limits may warrant a stricter default blast-radius cap than Meta—flagged as a technical spike, not a product guess.



One Thesis, Six Documents

Earn autonomy incrementally. Make the AI's boundaries explicit at every layer.

- Case Study Narrative
- User Stories & Acceptance Criteria
- Metrics & Attribution Spec
- Release Plan
- PRD
- AI Workflow Design
- Architecture & API Integration Doc